LEDGER · 2026-09-01 · POLYMER CHEMISTRY / STRUCTURAL COMPOSITES

High-Impact Polydicyclopentadiene (pDCPD) Resins via Frontal Ring-Opening Metathesis Polymerization (FROMP)

REJECTED

This concept did **not** clear the framework.

The reasoning is published in full below.

What the assessment found

Incumbent benchmark	Structural Epoxy Resin
Measured advantage	300%-400% (3-4x) improvement in ballistic energy absorption (KE50) and 50% smaller delamination area compared to structural epoxy
Time to first revenue	6 months
Gross margin at parity	60.7%
Startup CapEx	$134,500
Payback from first sale	1.9 months
Capital productivity	17.02x

Regulatory risk	Moderate — qualification costs reported ($85,000 estimated)

Why it failed

- Margin fails the operating-cost check

Assessment

Fails unit economics under Sub-Test D. At the incumbent selling price of $4.77/kg and unit OPEX of $1.86/kg, the gross margin is 61.0%, failing the strict 70% threshold required at price parity.

Also flagged

- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-01 against seven gates plus the voiding sub-tests. How the method works · Browse all concepts